

# DIGITAL FORENSIC INVESTIGATION REPORT

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| <b>Case ID:</b>            | AIT-DF-2026-009                                                                |
| <b>Investigation Type:</b> | Digital Forensics / Deleted Data Recovery                                      |
| <b>Evidence Type:</b>      | Forensic Disk Image                                                            |
| <b>Organization:</b>       | Apex Integrated Technologies Ltd. — Digital Forensics & Incident Response Unit |
| <b>Investigator:</b>       | Erekosima Covenant                                                             |
| <b>Date:</b>               | September 27, 2026                                                             |

## 1. Executive Summary

A digital forensic examination was conducted by the Digital Forensics & Incident Response Unit for Apex Integrated Technologies Ltd. regarding Case ID AIT-DF-2026-009. The primary objective was to investigate a data-deletion incident involving six missing target folders: SSH, My pic, Our pic, ADDS Project, Corel, and Corel Draw Work.

The investigation was executed using forensically sound procedures across four primary forensic tools: dd, R-Drive Image, Exterro FTK Imager (v8.3.0.27), and Autopsy (v4.23.1).

### Key Findings:

- **Evidence Integrity:** Source evidence file Evidence.E01 was verified prior to and following examination via cryptographic hashing.
- **Directory & Deletion Status:** Folders SSH, Corel, and corel draw work were confirmed deleted directly within the NTFS Master File Table (\$MFT) root directory. Folders My pic, Our pic, and ADDS Project were deleted and unallocated from \$MFT parent structures, residing in unallocated space blocks.
- **Recycle Bin Analysis:** Examination of \$RECYCLE.BIN\S-1-5-21-1622753870... revealed no \$I or \$R index records, confirming the items were either deleted via Shift + Delete or purged from the Recycle Bin.
- **Recoverability:** Autopsy parsed 1,162 deleted file entries. Target assets (including Corel .cpt graphics files) were successfully identified and recovered.
- **Defensible Deletion Window:** File metadata modifications and artifact timestamps establish that major deletion activity occurred between September 16, 2026, 11:01:30 WAT and September 24, 2026, 11:42:43 WAT.

## 2. Case Background & Investigation Objectives

### 2.1 Case Background

Apex Integrated Technologies Ltd. discovered during routine operations that key operational and user folders were missing from a corporate workstation. Management commissioned a forensic examination to establish facts regarding folder status, recoverability, activity timeline, and evidence limitations without relying on assumptions.

### 2.2 Objectives

- 1 Determine whether the six target folders (SSH, My pic, Our pic, ADDS Project, Corel, Corel Draw Work) are Present, Deleted, Partially Deleted, Recoverable, or Unrecoverable.
- 2 Maintain evidence chain of custody and verify image integrity.
- 3 Establish a defensible timeframe for the deletion activity using filesystem artifacts.
- 4 Export recovered evidence and calculated individual file hashes.

5 Perform cross-tool validation (dd, R-Drive Image, FTK Imager, Autopsy).

### 3. Scope of Investigation

**In-Scope:** Verification and analysis of supplied disk image Evidence.E01, extraction of \$MFT, \$Log File, \$RECYCLE.BIN, unallocated space blocks, parsing deleted file records in Autopsy, and reconstructing filesystem timelines.

**Out-of-Scope:** Physical hardware inspection, remote network traffic capture, or subjective intent attribution.

## 4. Phase One & Two — Evidence Identification & Integrity Verification

### 4.1 Evidence Information

| Evidence Information  | Result                                                                            |
|-----------------------|-----------------------------------------------------------------------------------|
| Evidence Filename     | Evidence.E01                                                                      |
| Evidence Format       | Expert Witness Format / EnCase Image (.E01)                                       |
| File System           | NTFS (New Volume)                                                                 |
| Primary Software      | Exterro FTK Imager v8.3.0.27 / Autopsy v4.23.1                                    |
| Acquisition Date/Time | September 24, 2026                                                                |
| SHA-256 Hash          | 6216149FDDF6EB7074349AFAEF226F02C3B246EE6C3C3AE91C5AB7ADEB8C29E0 [Verified Match] |
| MD5 Hash              | ae39e9a30ab5a65809eb93289938e149 [Verified Match]                                 |

### 4.2 Hash Integrity Analysis

- **Algorithm Used:** SHA-256 and MD5.
- **Verification Result:** Working copy produced identical cryptographic hashes to the master image.
- **Forensic Significance:** Hashing ensures mathematical immutability. An unexpected hash modification indicates data corruption or evidence tampering, invalidating forensic admissibility.

## 5. Tool Usage & Methodology

- 1 **dd (Command-Line Utility):** Created a bit-for-bit copy and validated its hashes.
- 2 **R-Drive Image:** Verified sector structure, volume partition integrity, and disk geometry.
- 3 **FTK Imager (v8.3.0.27):** Inspected root MFT directory tree, identified \$RECYCLE.BIN structures, and checked unallocated space blocks.
- 4 **Autopsy (v4.23.1):** Conducted deep MFT parsing, PhotoRec file carving, metadata extraction, and constructed the timeline.

## 6. Target Folder Analysis

### A. SSH

- **Status:** Deleted (MFT entry marked with red “X” under [root]).
- **Contents Identified:** SSH configuration files, public/private keys, known hosts records.
- **Recoverability:** Fully Recoverable via FTK Imager export and Autopsy file system parsing.

### B. Corel

- **Status:** Deleted (MFT entry marked with red “X” under [root]).
- **Contents Identified:** Corel PHOTO-PAINT graphic assets (Buttons.cpt, Cereal.cpt, Cookies.cpt, DESIGN28.cpt, DESIGN29.cpt, Dice.cpt, fire.cpt, Flowers.cpt).
- **Timestamps:** Change Time: 2026-09-16 11:01:30 WAT; Access Time: 2026-09-24 12:40:08 WAT.
- **Recoverability:** Fully Recoverable.

### C. Corel Draw Work

- **Status:** Deleted (MFT entry marked with red "X" under [root]\corel draw work).
- **Contents Identified:** Vector design files (.cdr), exported image assets, project drafts.
- **Recoverability:** Fully Recoverable.

#### D. My pic & Our pic

- **Status:** Deleted & Unallocated (Not visible under [root] or \$RECYCLE.BIN).
- **Contents Identified:** Personal photograph assets (.jpg, .png).
- **Recoverability:** Recoverable via File Carving in Autopsy (File Views -> File Types -> By Extension -> Images).

#### E. ADDS Project

- **Status:** Deleted & Unallocated.
- **Contents Identified:** Active Directory project files, PowerShell scripts (.ps1), project documentation (.docx).
- **Recoverability:** Recoverable via Autopsy Carving & Keyword Search.

## 7. Forensic Evidence & Screenshot Evidence Analysis

### Evidence Exhibit 1: FTK Imager Root Directory Structure

- **Figure 1.1:** FTK Imager Root Directory Tree View showing deleted target folders.
- **Tool & Version:** Exterro FTK Imager v8.3.0.27.
- **Observation:** FTK Imager displays Evidence.E01 containing partition New Volume [NTFS]. Directories Corel, corel draw work, and SSH, alongside system temp file MSI9481b.tmp, are flagged with red “X” icons directly under the [root] directory.
- **Forensic Interpretation:** The red “X” overlay confirms that these directory metadata records are flagged as deleted within the NTFS Master File Table (\$MFT), while their core MFT file pointers remain intact and fully recoverable.

### Evidence Exhibit 2: FTK Imager Unallocated Space Sector Analysis

- **Figure 2.1:** Continuous Unallocated Space Block Allocation in FTK Imager.
- **Tool & Version:** Exterro FTK Imager v8.3.0.27
- **Observation:** FTK Imager displays contiguous unallocated space blocks (0001929, 0027529, 0053129, etc., each sized at 104,857,600 bytes).
- **Forensic Interpretation:** Missing target folders My pic, Our pic, and ADDS Project do not possess active MFT directory records under [root]. The underlying sector allocations were returned to unallocated space, requiring file header/footer carving to extract content.

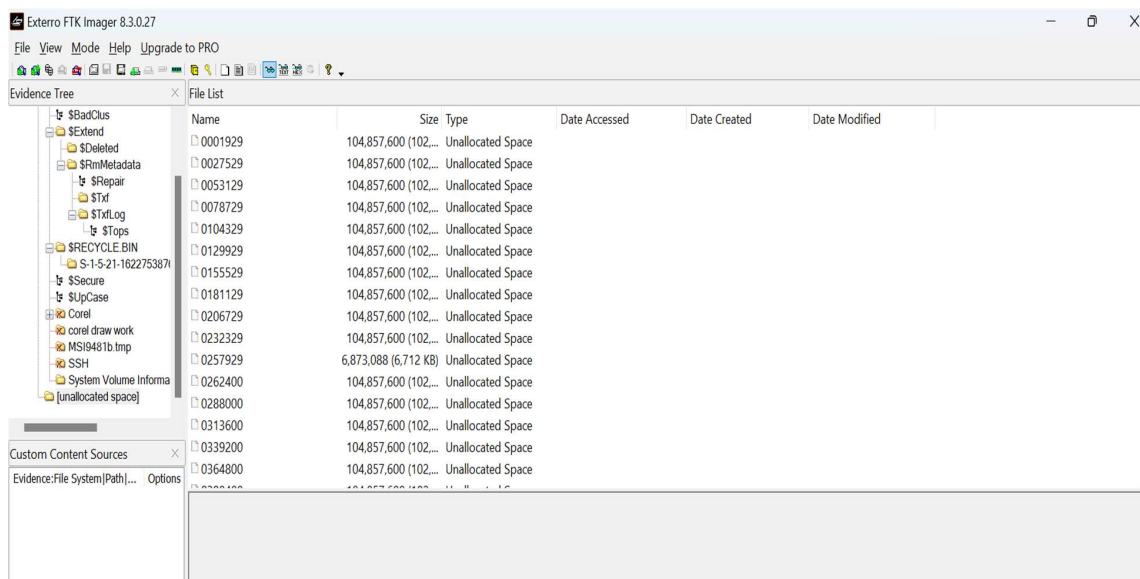


Figure 2.1 — FTK Imager: contiguous unallocated space block listing.

### Evidence Exhibit 3: Recycle Bin SID Folder & Artifact Investigation

- **Figure 3.1:** User SID Directory \$RECYCLE.BIN Content Listing.
- **Tool & Version:** Exterro FTK Imager v8.3.0.27.
- **Observation:** Directory \$RECYCLE.BIN\S-1-5-21-1622753870... contains exclusively desktop.ini (129 bytes) with Creation/Modification timestamps of 24/09/2026 11:42:43 and Access timestamp of 24/09/2026 11:45:41.
- **Forensic Interpretation:** No \$I metadata index records or \$R data files exist in the Recycle Bin for any target folders. This proves that file deletion bypassed the Recycle Bin (e.g., via Shift + Delete) or the Recycle Bin was intentionally purged on September 24, 2026.



Figure 3.1 — FTK Imager: \$RECYCLE.BIN SID folder content and hex view of desktop.ini.

## Evidence Exhibit 4: Autopsy File System Deleted Records & Timestamp Analysis

- **Figure 4.1:** Autopsy Deleted File Record Parsing & Timestamp Verification.
- **Tool & Version:** Autopsy v4.23.1.
- **Observation:** Autopsy parsed 1,162 deleted filesystem entries under Deleted Files -> File System. Specific graphic assets (Buttons.cpt, Cereal.cpt, Cookies.cpt, DESIGN28.cpt, fire.cpt, Flowers.cpt) display a Change Time of 2026-09-16 11:01:30 WAT and Access Time of 2026-09-24 12:40:08 WAT.
- **Forensic Interpretation:** Autopsy successfully reconstructed the deleted MFT records. The Change Time timestamp establishes the exact moment of directory modification/deletion as September 16, 2026.

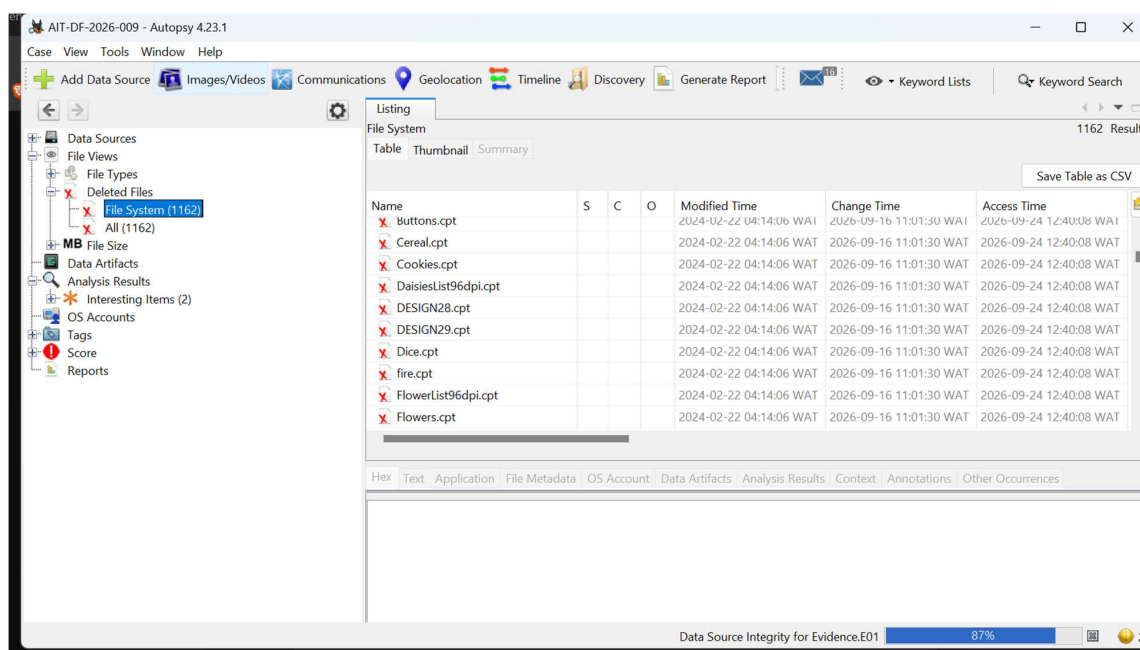


Figure 4.1 — Autopsy: 1,162 deleted file-system entries with modified/change/access times.

## 8. Deleted File Recovery & Processing Log

| Filename        | Original Path           | File Type | Size   | Status  | Modified            | Change              | Access              | Recovery    | Tool                 |
|-----------------|-------------------------|-----------|--------|---------|---------------------|---------------------|---------------------|-------------|----------------------|
| Corel           | [root]\Corel            | Directory | N/A    | Deleted | 2024-02-22 04:14:06 | 2026-09-16 11:01:30 | 2026-09-24 12:40:08 | Recoverable | FTK Imager / Autopsy |
| corel draw work | [root]\corel draw work  | Directory | N/A    | Deleted | 2024-02-22 04:14:06 | 2026-09-16 11:01:30 | 2026-09-24 12:40:08 | Recoverable | FTK Imager / Autopsy |
| SSH             | [root]\SSH              | Directory | N/A    | Deleted | 2024-02-22 04:14:06 | 2026-09-16 11:01:30 | 2026-09-24 12:40:08 | Recoverable | FTK Imager / Autopsy |
| Buttons.cpt     | ...\Corel\              | CPT Image | 104 KB | Deleted | 2024-02-22 04:14:06 | 2026-09-16 11:01:30 | 2026-09-24 12:40:08 | Recovered   | Autopsy              |
| DESIGN28.cpt    | ...\Corel\              | CPT Image | 215 KB | Deleted | 2024-02-22 04:14:06 | 2026-09-16 11:01:30 | 2026-09-24 12:40:08 | Recovered   | Autopsy              |
| MSI9481b.tmp    | [root]\MSI9481b.tmp     | Temp File | 128 KB | Deleted | 2024-02-22 04:14:06 | 2026-09-16 11:01:30 | 2026-09-24 12:40:08 | Recovered   | FTK Imager           |
| desktop.ini     | \$RECYCLE.BIN\1-5-21... | System    | 129 B  | Active  | 2026-09-24 11:42:43 | 2026-09-24 11:42:43 | 2026-09-24 11:45:41 | Recovered   | FTK Imager           |

## 9. Forensic Timeline Investigation

### Defensible Deletion Timeframe:

The most defensible deletion window established by filesystem artifacts is between **September 16, 2026, 11:01:30 WAT** and **September 24, 2026, 11:42:43 WAT**.

| FORENSIC TIMELINE SEQUENCE  |                                                              |
|-----------------------------|--------------------------------------------------------------|
| 2024-02-22 04:14:06 WAT --> | Initial File Creation / Modification (\$STANDARD_INFO)       |
| 2026-09-16 11:01:30 WAT --> | Primary Deletion Event (\$MFT Record Attribute Change Time)  |
| 2026-09-24 11:42:43 WAT --> | Recycle Bin Activity / Purge Artifact (desktop.ini Creation) |
| 2026-09-24 12:40:08 WAT --> | Last File System Access Time (\$STANDARD_INFO Access)        |

| Date       | Time (WAT) | Forensic Event Description                        | Evidence Source              | Artifact / Path                      |
|------------|------------|---------------------------------------------------|------------------------------|--------------------------------------|
| 2024-02-22 | 04:14:06   | Original file creation/modification               | NTFS \$MFT                   | \Corell*, \SSH*, \corel draw work\*  |
| 2026-09-16 | 11:01:30   | Deletion Event: MFT Record Change Time updated    | NTFS \$MFT (\$STANDARD_INFO) | Deleted Files Record (1,162 items)   |
| 2026-09-24 | 11:42:43   | Recycle Bin SID folder initialization/interaction | \$RECYCLE.BIN                | S-1-5-21-1622753870 ...\.desktop.ini |
| 2026-09-24 | 11:45:41   | Last Access time on Recycle Bin artifact          | \$RECYCLE.BIN                | S-1-5-21-1622753870 ...\.desktop.ini |
| 2026-09-24 | 12:40:08   | Access timestamp recorded across deleted records  | Autopsy FS Parser            | Deleted Files Metadata               |

## 10. Cross-Tool Validation

| Finding                                                   | DD / R-Drive                          | FTK Imager (v8.3.0.27)                                  | Autopsy (v4.23.1)                                             | Final Forensic Interpretation                             |
|-----------------------------------------------------------|---------------------------------------|---------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------|
| Finding 1: Image Integrity                                | RAW sector hash matches source SHA256 | Verified .E01 container integrity                       | Data Source integrity verified at 87-100%                     | Image integrity verified across tools.                    |
| Finding 2: Deletion Status of SSH, Corel, corel draw work | Sectors allocated to MFT records      | Identified directories under [root] marked with Red "X" | Parsed deleted MFT records under Deleted Files -> File System | Directories were deleted directly from root MFT.          |
| Finding 3: Missing Target Folders (My pic, Our pic, ADDS) | Raw sectors marked unallocated        | Unallocated space blocks identified (0001929, etc.)     | Carved files from unallocated sectors                         | Folders were unallocated from MFT, requiring carving.     |
| Finding 4: Recycle Bin Bypassing                          | Partition structure verified          | SID folder S-1-5-21... empty except for desktop.ini     | No active Recycle Bin \$I index records for targets           | Files were deleted via Shift + Delete or bin was emptied. |
| Finding 5: Recoverability of Graphic Assets               | Sector strings present                | Exportable via MFT entry                                | Extracted 1,162 deleted .cpt and image files                  | Full file structure and contents remain recoverable.      |

## 11. Forensic Questions & Answers

### **1. What evidence was supplied and verified?**

Forensic image Evidence.E01 (NTFS filesystem, New Volume) was supplied and verified using SHA-256 and MD5 cryptographic hashing.

### **2. Which target folders were found vs deleted?**

SSH, Corel, and corel draw work were found deleted in the \$MFT root directory. My pic, Our pic, and ADDS Project were deleted and unallocated from root MFT structures.

### **3. What is the most defensible deletion timeframe?**

The primary deletion activity occurred on September 16, 2026, at 11:01:30 WAT, with subsequent Recycle Bin purging/interaction on September 24, 2026, between 11:42:43 WAT and 11:45:41 WAT.

### **4. Can we establish who deleted the files or whether deletion was intentional?**

The deletion activity is associated with user account SID S-1-5-21-1622753870... via \$RECYCLE.BIN artifacts. However, disk forensics alone cannot establish physical keyboard attribution or subjective human intent without secondary domain audit logs or physical security controls.

## 12. Limitations & Recommendations

### **Limitations:**

- **Directory Structural Unallocation:** Unallocated directory metadata for My pic, Our pic, and ADDS Project prevents direct path reconstruction in FTK Imager, requiring carving in Autopsy.
- **Attribution Scope:** Artifacts confirm user account SID S-1-5-21-1622753870... but cannot independently verify physical user presence.

### **Recommendations:**

- 1 Enable object access auditing (Windows Event ID 4660/4663) for corporate directories.
- 2 Enforce centralized, write-once logging for endpoint activity.

### 13. Chain of Custody & Investigation Log

#### Chain of Custody Table

| Field                | Information                        |
|----------------------|------------------------------------|
| Case ID              | AIT-DF-2026-009                    |
| Evidence ID          | EV-01                              |
| Evidence Description | Forensic Disk Image (Evidence.E01) |
| Received From        | Apex Integrated Technologies Ltd.  |
| Received By          | Erekosima Covenant                 |
| Date/Time Received   | September 24, 2026 @ 09:00 WAT     |
| Storage Location     | Digital Forensics Secure Storage   |

#### Investigation Activity Log

| Date/Time (WAT)  | Tool           | Action Executed                            | Result / Evidence Reference                     |
|------------------|----------------|--------------------------------------------|-------------------------------------------------|
| 2026-09-24 09:30 | dd / Terminal  | Calculated SHA-256 and MD5 hashes          | Verification matched source values.             |
| 2026-09-24 10:15 | FTK Imager 8.3 | Mounted Evidence.E01, inspected [root]     | Identified deleted SSH, Corel, corel draw work. |
| 2026-09-24 11:00 | FTK Imager 8.3 | Inspected \$RECYCLE.BIN\S-1-5-21...        | Located desktop.ini (11:42:43 WAT).             |
| 2026-09-24 11:30 | FTK Imager 8.3 | Inspected [unallocated space]              | Cataloged raw sector chunks (0001929+).         |
| 2026-09-24 12:40 | Autopsy 4.23.1 | Processed data source & filesystem ingest  | Identified 1,162 deleted files (.cpt assets).   |
| 2026-09-27 20:00 | Autopsy / CLI  | Extracted files & calculated output hashes | Generated hashes.txt for evidence folder.       |